JACK J. BURLESON

MACHINE LEARNING • DATA SCIENCE • AI ENGINEERING & SAFETY

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PROFESSIONAL SUMMARY

UW–Madison M.S. candidate in Psychology: Data Science in Human Behavior (expected Dec 2026); B.S. in Psychology (neuroscience concentration, sociology minor), UW–Parkside. Applied AI practitioner building full-lifecycle systems spanning multi-agent document-intelligence pipelines, transformer fine-tuning and calibration, LLM evaluation harnesses, computer-vision classification, and behavioral prediction models. Program capstone with American Family Insurance as industry partner, developed through biweekly working sessions with their Enterprise Data Science lead. Five years of high-volume operations experience in quality assurance, documentation, and process improvement.

EDUCATION

University of Wisconsin–Madison *Madison, WI* August 2025 – December 2026

M.S. Candidate – Psychology: Data Science in Human Behavior · **GPA: 3.85**

- Graduate program in which roughly 95% of coursework is data science and data engineering — building on an undergraduate foundation in neuroscience research, experimental design, and statistical analysis.
- Advanced training in machine learning (neural networks, regularized and ensemble models, transformer fine-tuning and LLM evaluation), statistical modeling, and large-scale data systems (Spark, gRPC services, HDFS, Parquet, SQL, Docker).
- Industry capstone with American Family Insurance (see Projects).

University of Wisconsin–Parkside *Kenosha, WI* September 2020 – May 2025

B.S., Psychology

Concentration: Neuroscience. **GPA: 3.93**

- Coursework spanning cognitive and computational neuroscience, psychological theory, clinical psychology, psychopharmacology, psychometrics, psychological statistics, and early data science preparation — plus philosophy, neuroethics, and pre-law.
- Minor in Sociology; Provost's List (7x), Dean's List (8x), Distinguished Achievement Award (2025), and Psychological Research Certification (300 lab hours).

PROFESSIONAL EXPERIENCE

UW–Madison Recreation & Wellbeing — Building Services *Madison, WI* Jun 2026 – Present

- Maintain recreation and wellbeing centers across campus to high standards of cleanliness, safety, and operational readiness.
- Respond to facility and equipment issues across multiple locations, keeping centers safe and fully operational during peak use.

UW–Parkside — Undergrad Researcher, Cognitive Neuroscience Lab *Kenosha, WI* Jun 2024 – Aug 2025

- Conducted cognitive neuroscience research under Dr. Melissa Gregg, designing and pilot-testing an experimental study on truth, deception, and cognitive workload (study code: github.com/Exios66/truth-deception-architecture).
- Operated EEG systems and analyzed neural time-series data to quantify brain activity, contributing signal-processing and statistical insights to ongoing lab research on sleep quality and cognitive load.
- Presented findings at the 2025 Midwestern Psychological Association conference in Chicago and at Research in the Rotunda.

Meijer Distribution — Logistics Coordinator *Pleasant Prairie, WI* Jul 2020 – Aug 2025

- Promoted from Retail Loader to Logistics Coordinator for reliability in quality assurance and data-driven process documentation across high-volume operations.
- Analyzed quality-assurance data for 64 daily deliveries across a Midwest regional network, maintaining error rates below 1% through root-cause investigation and process refinement.
- Partnered with warehouse and transportation teams to investigate quality problems and resolve logistics exceptions before shipment.

PROJECTS

Digital-Mailroom *Data Science in Human Behavior capstone — American Family Insurance (industry partner)*

- Capstone developed with American Family Insurance as industry partner: 5 Python packages addressing document-intelligence problems scoped by their Enterprise Data Science division — entity extraction, classification, and scalable legal-document workflows.
- Implemented a unified LangGraph pipeline integrating legal-document routing, prompt experimentation, deterministic scoring, corpus analysis, and deployment workflows in a reproducible Docker environment.

Mailroom-ML *ML training, evaluation & fine-tuning*

- Developed a ModernBERT document classifier with end-to-end training, evaluation, and fine-tuning pipelines — including data lineage tracking, probability calibration, and production routing logic.
- Implemented PyTorch training pipelines with Transformers and Hugging Face datasets, SciPy-based evaluation, and ONNX export for FastAPI serving on Modal; maintained pytest coverage throughout.

RVL-CDIP-Classifier *Computer vision & prompt engineering*

- Developed a Python toolkit that converts TIFF/PDF pages to 300-DPI images, runs OCR, and classifies documents into 16 RVL-CDIP categories with OpenRouter vision models.
- Tracked 29 prompt versions with confusion matrices, misclassification analyses, and cost projections to guide prompt and model selection.

Psychometric Digital Twin *Behavioral ML research · UW–Madison*

- Built a behavioral prediction pipeline on 241 complete respondents from 252 matched survey records — Ridge, Elastic Net, k-NN, Random Forest, XGBoost, and MLP predicting communication-apprehension scores from demographic, employment, geographic, and transit features, with cross-validation and SHAP feature importance.
- Validated the PRCA's psychometric structure with factor analysis, then built LLM 'digital twin' personas via tiered embodiment prompts (Ollama/OpenRouter/vLLM) and benchmarked them against the ML baselines on shared metrics (MAE, band accuracy, F1) — analyzing systematic bias by demographic group.

Neuron Atlas *Mechanistic interpretability · live browser model*

- Built a browser interpretability tool that maps all 8,192 MLP neurons of TinyStories-8M and supports live steering, ablation and clamping beside random-direction controls; recorded Qwen studies extend it beyond live browser inference.
- Implemented a TypeScript GPT-Neo forward pass with a KV cache, a Three.js atlas and a Python sweep/analysis pipeline, verified against PyTorch and covered by Playwright tests for navigation, replay, phones and failed downloads.

MapReduce-Reloaded *Distributed systems · Go + Python*

- From-scratch Go re-implementation of Google's MapReduce (Dean & Ghemawat, OSDI 2004), extended for modern infrastructure: gRPC coordinator/worker protocol, Docker Compose cluster deployment, and a Python reference implementation sharing the same protobuf contract and file format.

TECHNICAL SKILLS

Languages & Data	Python (pandas, NumPy, scikit-learn, PyTorch, Hugging Face, PyArrow), R, MATLAB, SQL, Parquet, Hive, JavaScript/TypeScript, Git
Machine Learning	transformer fine-tuning, neural networks, regularization & penalized models (Lasso/Ridge), ensemble methods (random forests, bagging), resampling methods, model evaluation & calibration, cross-validation, SHAP/feature importance, error analysis, computer vision
AI Engineering	LangGraph (LangChain), multi-agent pipelines, LLM eval harnesses (incl. bias & robustness evals), Prompt Engineering, model serving (vLLM, Ollama, LM Studio, OpenRouter), ONNX + FastAPI deployment on Modal
Engineering & Cloud Services	Docker, gRPC, Apache Spark, PySpark, HDFS, shell scripting (bash), pytest, Quarto, CHTC (high-throughput computing), AWS, Azure, GCP